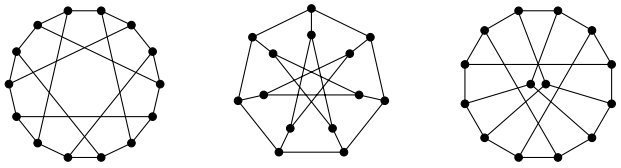
第一章　图论初步

1.1　试卷: 图论初步

原网页

24 秋图论初步期末试卷 (吴河辉)

试题共八道题，其中前四题每题 10 分，后面四题每题 20 分，选三题作答，并在上面方框中划去不做的题目。



1. Which pair of graphs below are isomorphic? _____
2. How many isomorphism classes of 1204-vertex trees with diameter 3? _____
3. What is the minimum size of a maximal matching in the cycle C_{2024}? _____
4. What is the number of faces in a 100-vertex plane graph with average degree 3.14? _____
5. Let G be an r-connected graph of even order having no $K_{1,r+1}$ as an induced subgraph. Prove that G has a 1-factor.
6. Prove that every subgraph of $K_{n,n}$ having at least $n(n-1)+2$ edges has cycles of all even lengths from 4 through $2n$.
7. Prove that every triangle-free graph with at most $\frac{t^2+3t-2}{2}$ vertices is t-colorable.
8. Let G be a graph with $\delta(G)=3$ and $\bar{d}(G)<\frac{10}{3}$. Prove that G has a 3-vertex whose neighbors have degree-sum at most 10.

23 秋图论初步期末试卷 (吴河辉)

试题在六道题里选做五题，并在答题纸上划去不做的题目。

1. For $k>2$, prove that if the vertices of a k-regular graph partitions into sets A and B that each induces a tree, then $|A|=|B|$.
2. Prove that every nontrivial tree has a maximum matching that covers every non-leaf vertex plus at least one leaf.
3. Prove or disprove: if G is a 2-connected graph that is not a cycle, then it contains a cycle C such that $G-V(C)$ is connected.
4. Place n vertices around a circle. Let G_n be the 4-regular graph obtained by making each vertex adjacent to the two nearest vertices in each direction. For $n\geq 5$, prove that G_n is the union of two

Hamiltonian cycles.

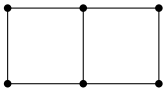
5. A **parity edge-coloring** of a graph G is an edge-coloring such that on every path, some color appears an odd number of times, let $p(G)$ denote the minimum number of colors in a parity edge-coloring of G. Show that $p(K_5) = 7$.
6. Prove that every plane triangulation with minimum degree 5 has two 3-faces sharing an edge such that the non-shared vertices have degree-sum at most 11. (Hint: Use Vertex discharging, and each 5-vertex get $1/5$ charge from each neighbor first.)

22 秋图论初步期末试卷 (吴河辉)

试题在六道题里选做五题，并在答题纸上标明不做的题目。

1. Let $E(A, V - A)$ be an edge cut of a graph G with even size, and v is an odd vertex of A, show that there is another odd vertex u in A.
2. Let x, y be two vertices in G with $\kappa'(x, y) = 2k$ for some integer k. Let X be the set of odd vertices in G and $|X| = 2l$. Show that there exists a disjoint pairing $\{(s_i, t_i) : 1 \leq i \leq l\}$ of X, together with l edge-disjoint paths $\{P_i : 1 \leq i \leq l\}$, where P_i joins s_i and t_i for each i, such that after removing the edges $\bigcup_{i=1}^{l} E(P_i)$ from G, x and y are still $2k$ edge connected.
3. A mouse eats its way through a $3 \times 3 \times 3$ cube of cheese by eating all the $1 \times 1 \times 1$ subcubes. If it starts at a corner subcube and always moves on to an adjacent subcube (sharing a face of area 1), can it do this and eat the center subcube last? Give a method or prove impossible. (Ignore gravity.)
4. Recall that a graph is Class 1 if $\chi'(G) = \Delta(G)$. Let G be a Class 1 regular graph of even degree. Prove that $|E(G)|$ is even.
5. The surface of a football is a planar graph with only 5-faces and 6-faces; Use graph theory to answer the following question (No credit without a proof).
   (a) How many 5-faces of it?
   (b) It is the smallest such planar graph with no adjacent 5-faces. How many vertices, faces and edges of it?
6. Prove that a graph is k-colorable if every vertex lies in fewer than $\binom{k}{2}$ odd cycles. (Hint: Use induction on $|V(G)|$).

21 秋图论初步期末试卷 (吴河辉)

1. For $k \in \mathbb{N}$, prove that every graph G has a k-partite subgraph H with at least $(1 - 1/k)|E(G)|$ edges.
2. Given a graph G, suppose that $S \subseteq V(G)$ is covered by some matching. Prove or disprove each statement below:
   (a) S is covered by some maximum matching.
   (b) S is covered by every maximum matching.
3. Let G be a 3-regular graph whose edge set has exactly one partition into three perfect matchings. Prove that G is Hamiltonian.
4. Prove that the graph on the right is not 2-choosable. 

5. Prove or disprove: If G is a k-edge-connected graph with minimum degree greater than k, then some two vertices in G are joined by more than k-edge-disjoint paths.
6. Let G be a planar graph with minimum degree 5. Let $e_{i,j}$ be the number of edges with endpoints of degrees i and j. Prove that $e_{5,6} \geq 60$ when $e_{5,5} = 0$.